UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

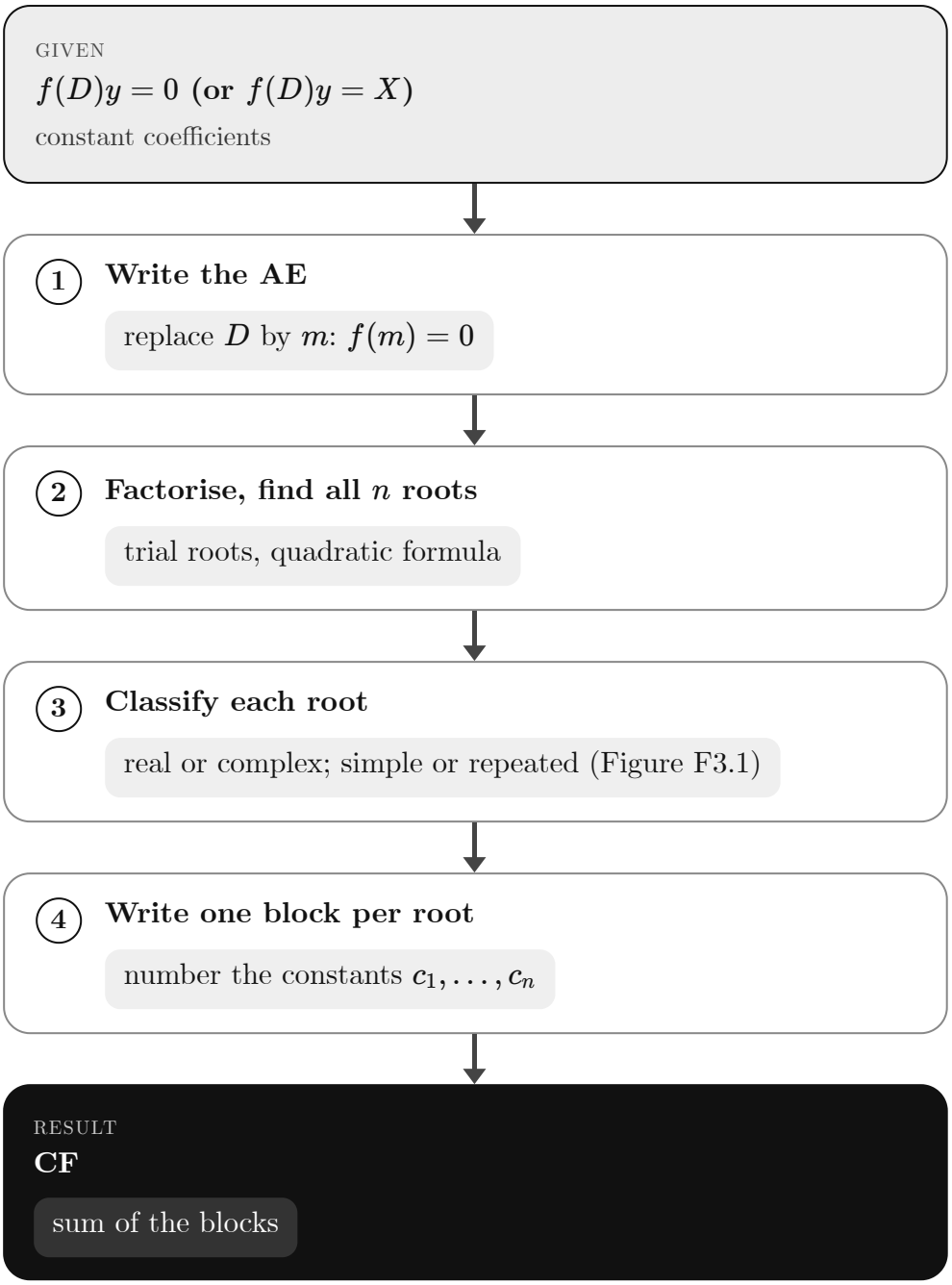
Class 002 · Solutions of homogeneous DE: complementary functions · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Unit 2 deck 002, slide 7 (rendered .pptx)
T002 00:09:10	Unit 2 AV-summary transcript 002 (“Rules for finding the Complementary Function, Problems”), segment start
CW U2 P8	Unit 2 class-work problem 8
ESA Jan-24 s4 2c	Past paper: UE23MA141A Jan 2024 set 4, question 2.c

Contents

1. Figure F1.1 · Finding the complementary function
2. Figure F2.1 · Which case? A second-order equation
3. Figure F3.1 · Which CF block does each root give?
4. Figure F4.1 · Constants from initial conditions (CW P8)
5. The three second-order cases
6. The five order-n cases of slides 002·11–13
7. Class-work problems 1–9 at a glance
8. The CF inside every distinct Unit 2 ESA question

1 The auxiliary equation and the complementary function

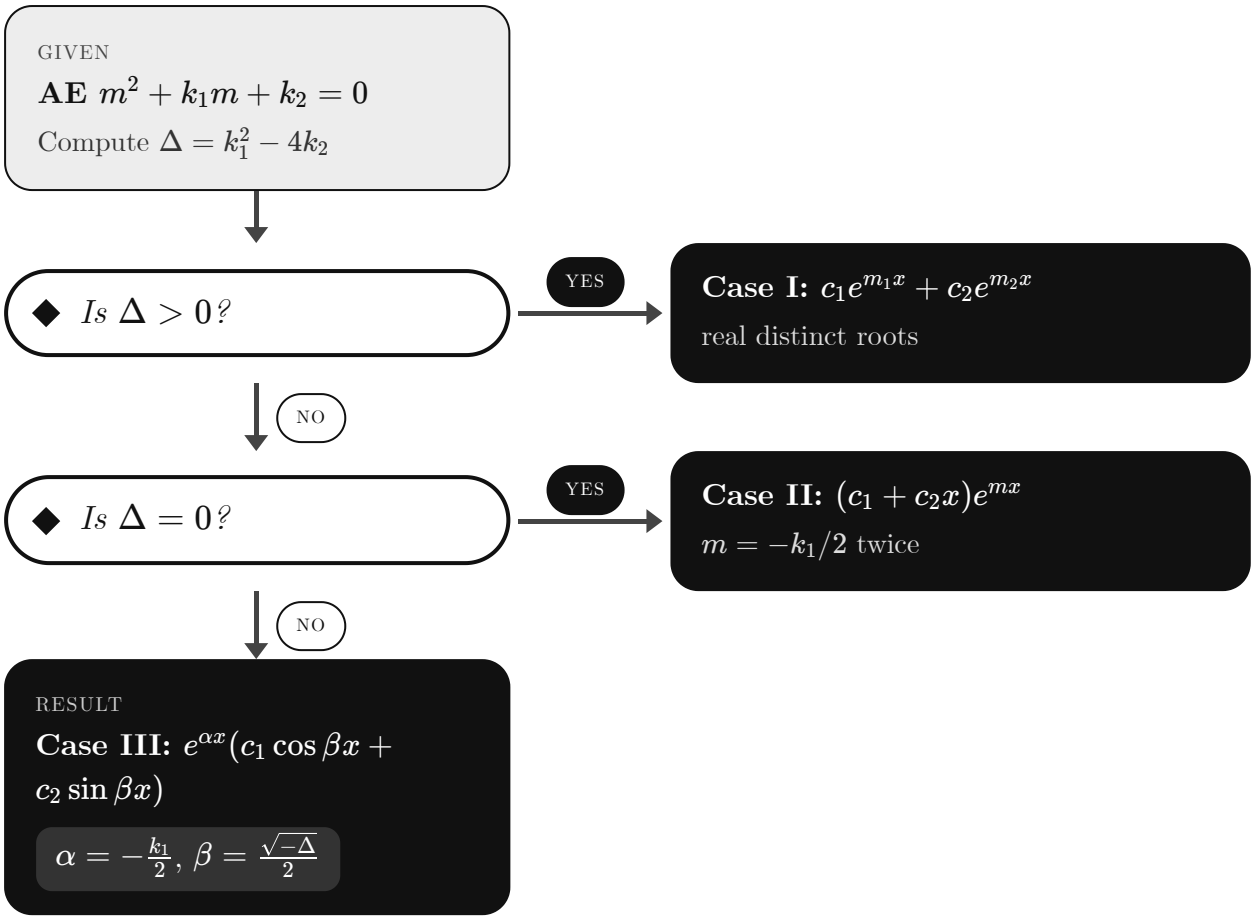


Remark. Check: exactly n constants for an n-th order equation. For a non-homogeneous equation this CF is half of the answer: $y = \mathbf{CF} + \mathbf{PI}$.

Figure F1.1. Finding the complementary function ESA ★★★

T002 00:03:18 · T002 00:07:40 · T002 00:08:07 · T002 00:20:41

2 Second-order equations: three cases



Remark. The discriminant test is added for clarity; the slides classify the roots after solving the AE.

Figure F2.1. Which case? A second-order equation ESA ★★★

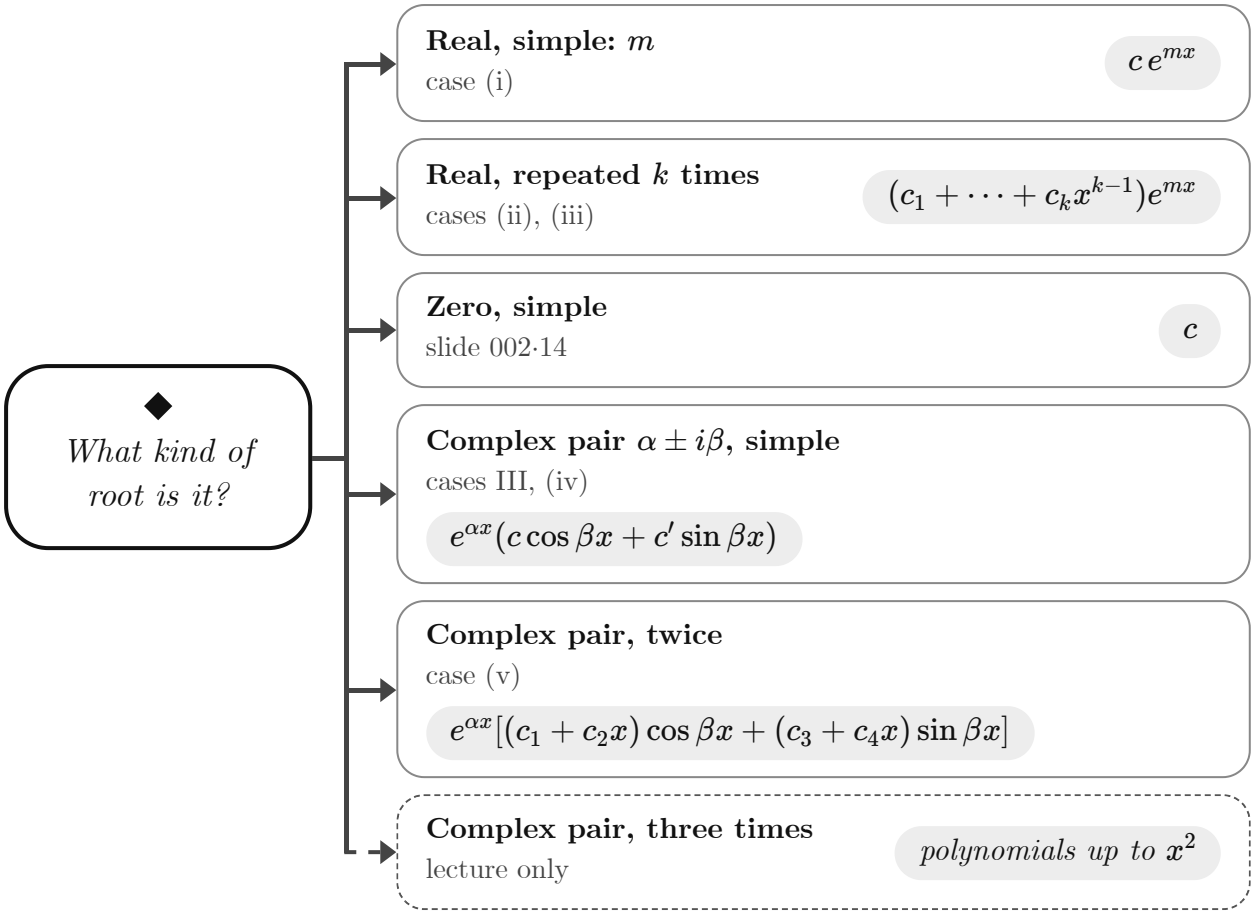
ADDED FOR CLARITY S002·7 · S002·8 · S002·9 · T002 00:11:56

The three second-order cases

Roots of AE	CF	Slide example	Answer	Shape (Figure G2.1)
real, distinct $m_1 \neq m_2$	$c_1 e^{m_1 x} + c_2 e^{m_2 x}$	$y'' + 2y' - 3y = 0$	$c_1 e^{-3x} + c_2 e^{x}$	sum of exponentials
real, equal m	$(c_1 + c_2 x)e^{mx}$	$y'' + 4y' + 4y = 0$	$(c_1 + c_2 x)e^{-2x}$	no oscillation
complex $\alpha \pm i\beta$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$	$y'' - 4y' + 13y = 0$	$e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$	oscillates; envelope $e^{\alpha x}$

S002·7 · S002·8 · S002·9

3 Equations of order n: combining the cases



Remark. Add the blocks for all the roots. The dashed route is from the lecture (T002 00:19:14), not the slides.

Figure F3.1. Which CF block does each root give? ESA ★★★

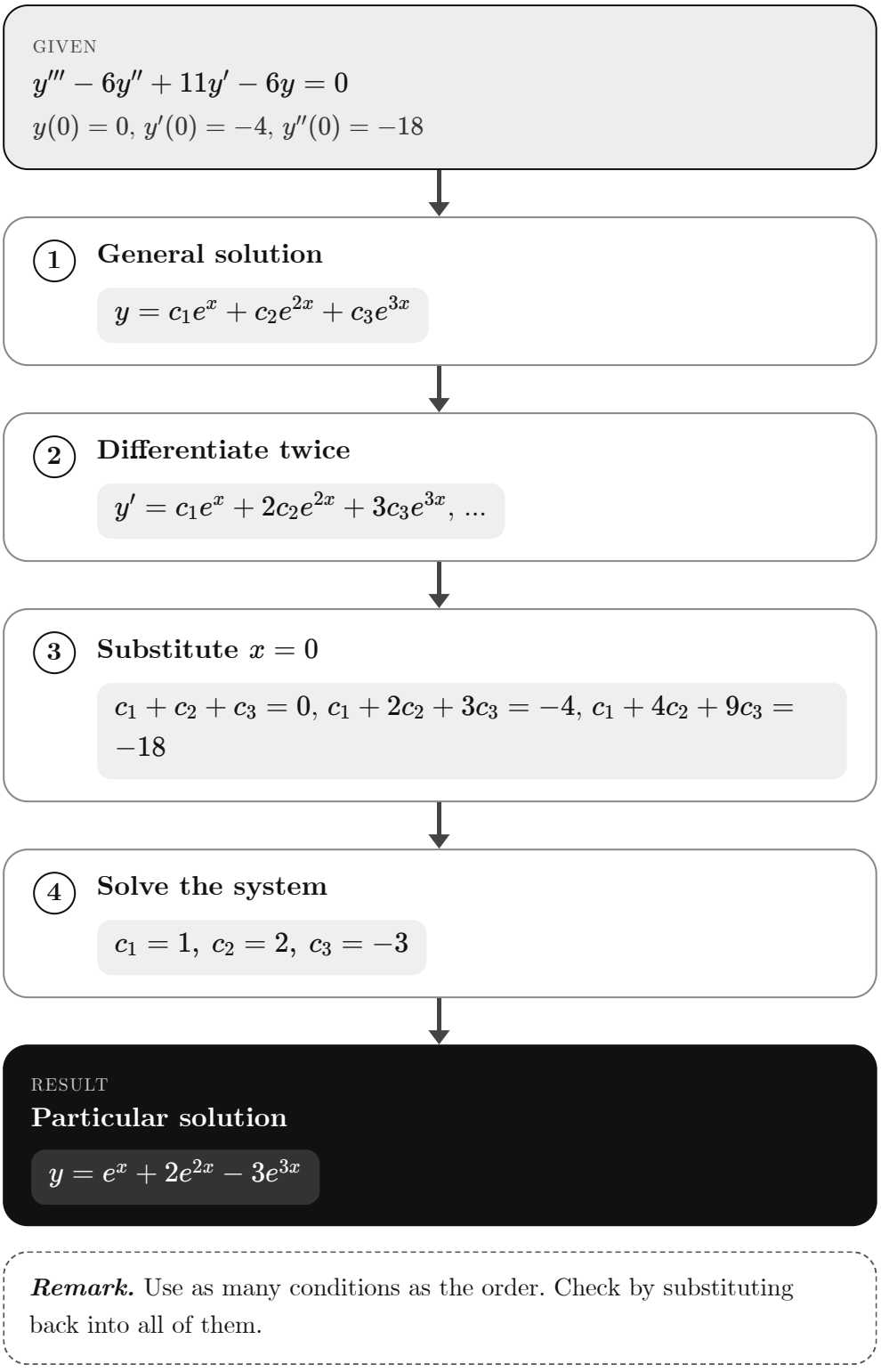
S002·11 · S002·12 · S002·13 · S002·14 · T002 00:19:14

The five order-n cases of slides 002·11–13

Case	Roots	General solution
(i)	all real, distinct	$c_1 e^{m_1 x} + \cdots + c_n e^{m_n x}$
(ii)	all real, equal to m	$(c_1 + c_2 x + \cdots + c_n x^{n-1})e^{mx}$
(iii)	four equal to m, rest distinct	$(c_1 + c_2 x + c_3 x^2 + c_4 x^3)e^{mx} + c_5 e^{m_5 x} + \ldots$
(iv)	$\alpha \pm i\beta$, $\gamma \pm i\delta$, rest distinct	$e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + e^{\gamma x}(c_3\cos\delta x + c_4\sin\delta x) + \ldots$
(v)	$\alpha \pm i\beta$ twice, rest distinct	$e^{\alpha x}[(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x] + \ldots$

S002·11 · S002·12 · S002·13

4 Solving problems, with initial conditions



Remark. Use as many conditions as the order. Check by substituting back into all of them.

Figure F4.1. Constants from initial conditions (CW P8) ESA ★★

CW U2 P8

Class-work problems 1–9 at a glance

CW	Roots	Case	Answer (verified)
P1	$6, -2$	distinct	$c_1 e^{6x} + c_2 e^{-2x}$
P2	$-2, -2$	equal	$(c_1 + c_2 x)e^{-2x}$
P3	$2 \pm i$	complex + IVP	$e^{2x}(2\cos x - 5\sin x)$
P4	$\frac{3}{2}, \frac{1}{2}$	distinct + IVP	$\frac{1}{2}(5e^{3x/2} - 3e^{x/2})$
P5	$1 \pm i,\ -1 \pm i$	two complex pairs	$e^{-t}(c_1\cos t + c_2 \sin t) + e^{t}(c_3 \cos t + c_4 \sin t)$
P6	$-1, 2, \frac{3}{2}, -\frac{1}{2}$	distinct	$c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x/2} + c_4 e^{-x/2}$
P7	$\pm 5i$ twice	repeated complex	$(c_1 + c_2 x)\cos 5x + (c_3 + c_4 x)\sin 5x$
P8	$1, 2, 3$	distinct + IVP	$e^{x} + 2e^{2x} - 3e^{3x}$
P9	$\pm 2i, \pm 3i$	complex + BVP	$2\sin 2x + \sin 3x$

CW U2 P1 · CW U2 P5 · CW U2 P9

The CF inside every distinct Unit 2 ESA question

Paper, Q	AE (after substitution)	Roots	CF
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(m-2)^2$	$2, 2$	$(c_1 + c_2 x)e^{2x}$
Dec-23 3b	$\theta^2 + 1$, $z = \log(1+x)$	$\pm i$	$c_1 \cos z + c_2 \sin z$
Dec-23 3c	$m^2 + 3m + 2$	$-1, -2$	$c_1 e^{-x} + c_2 e^{-2x}$
Jul-23 3a (= May-24 3a)	$4(\theta-2)^2$, $z = \log(2x+1)$	$2, 2$	$(c_1 + c_2 z)e^{2z}$
Jul-23 3b	$m^2 - 5m + 6$	$2, 3$	$c_1 e^{2x} + c_2 e^{3x}$
Jul-23 3c (= Jun-24 3a)	$(m+3)^2$	$-3, -3$	$(c_1 + c_2 x)e^{-3x}$
May-24 3c	$(m-1)^2$	$1, 1$	$(c_1 + c_2 x)e^{x}$
Jan-24 s1 2d	$m^2 + 4$	$\pm 2i$	$c_1 \cos 2x + c_2 \sin 2x$
Jan-24 s1 3a	$m^2 - 2m + 2$	$1 \pm i$	$e^{x}(c_1 \cos x + c_2 \sin x)$
Jan-24 s1 3c	$4(\theta-3)(\theta+1)$, $z = \log(2x+1)$	$3, -1$	$c_1 e^{3z} + c_2 e^{-z}$
Jan-24 s2 2d	$(m-1)(m-2)(m-3)$	$1, 2, 3$	$c_1 e^{x} + c_2 e^{2x} + c_3 e^{3x}$
Jan-24 s2 3a	$m^2 + 2m + 2$	$-1 \pm i$	$e^{-x}(c_1 \cos x + c_2 \sin x)$
Jan-24 s2 3c	$(\theta-1)^2$, $z = \log(x+2)$	$1, 1$	$(c_1 + c_2 z)e^{z}$

Paper, Q	AE (after substitution)	Roots	CF
Jan-24 s3 2c	$m(m-1)^2$	$0, 1, 1$	$c_1 + (c_2 + c_3 x)e^x$
Jan-24 s3 3b (LCR)	$m^2 + 40m + 625$	$-20 \pm 15i$	$e^{-20t}(c_1 \cos 15t + c_2 \sin 15t)$
Jan-24 s3 3d (= s4 3c)	$(\theta - 2)(\theta - 3)$, $z = \log x$	$2, 3$	$c_1 x^2 + c_2 x^3$
Jan-24 s4 2c	$(m-1)^2(m+1)^2$	$1, 1, -1, -1$	$(c_1 + c_2 x)e^x + (c_3 + c_4 x)e^{-x}$
Jan-24 s4 3b (LCR)	$m^2 + 20m + 500$	$-10 \pm 20i$	$e^{-10t}(c_1 \cos 20t + c_2 \sin 20t)$
Jun-24 3b	$2(\theta - 1)(2\theta + 1)$, $z = \log(2x - 1)$	$1, -\frac{1}{2}$	$c_1 e^z + c_2 e^{-z/2}$

ADDED FOR CLARITY

ESA Dec-23 3a · ESA Dec-23 3b · ESA Dec-23 3c · ESA Jul-23 3a · ESA Jul-23 3b · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s1 2d · ESA Jan-24 s1 3a · ESA Jan-24 s1 3c · ESA Jan-24 s2 2d · ESA Jan-24 s2 3a · ESA Jan-24 s2 3c · ESA Jan-24 s3 2c · ESA Jan-24 s3 3b · ESA Jan-24 s3 3d · ESA Jan-24 s4 2c · ESA Jan-24 s4 3b · ESA Jun-24 3b